First Year Sim — Hackathon Project Proposal

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Project summary

First Year Sim is a short mobile game that lets students rehearse their first university semester before it starts. Players manage money, marks and wellbeing through the real crises that push South African first-years out. A built-in Semester Forecast analyses each player's finances and choices to predict their riskiest weeks, and the game ends with a personal survival plan for their own institution.

One-line pitch: Every first-year crisis is predictable, so let students rehearse it.

Category: Access to higher education

> How might technology help young South Africans find suitable study pathways, compare funding options, complete applications and remain supported from acceptance through graduation?

Focus areas addressed

Focus area	Fit	How First Year Sim addresses it
Academic persistence	Primary	Prepares students for the shocks that cause first-year dropout
Student wellness	Strong	Wellbeing meter, stress events, and introduces counselling before it's needed
Cost planning	Strong	Allowance, rent, transport and family requests, ending in a real budget
NSFAS guidance	Partial	Late allowances and progression rules, as they affect the first year
Connectivity and accessible learning	Design	Low-data web app, no download, local languages

Pilot institution: University of Pretoria, Hatfield campus.

The problem

Getting accepted is not the hard part; surviving the first year is. Large numbers of South African students leave in their first year, and the crises that push them out are predictable.

The scale. Of the 917,092 students NSFAS funded in 2024, 780,125 were first years. Only about 130,000 of roughly 620,000 funded first years in 2023 were still funded in 2024. Some found bursaries or stopped qualifying, but many dropped out or did not meet the academic rules to continue (GroundUp via AllAfrica). Fundi's CEO has claimed that as many as 60% of first-years leave after their first term (EWN).

The causes. A study of NSFAS-funded students in Northern Gauteng found dropout linked to insufficient funding, late allocation of funds, poor communication, and late or missing allowance payments, alongside personal, institutional and health factors (Unisa repository). Academic pressure, emotional adjustment and pressure from home to contribute money also play a part (EWN).

Why it's predictable. Every year, some allowances arrive late, rent is due before the money, assignments cluster in the same weeks, and families ask for help. First-years meet all of it at once, for the first time, with no practice.

Support exists, but students find it too late. At UP, for example, food support, free counselling, first-year mentorship and academic coaching are all available (UP news). Many students only learn about them after they have already fallen behind.

Who we serve

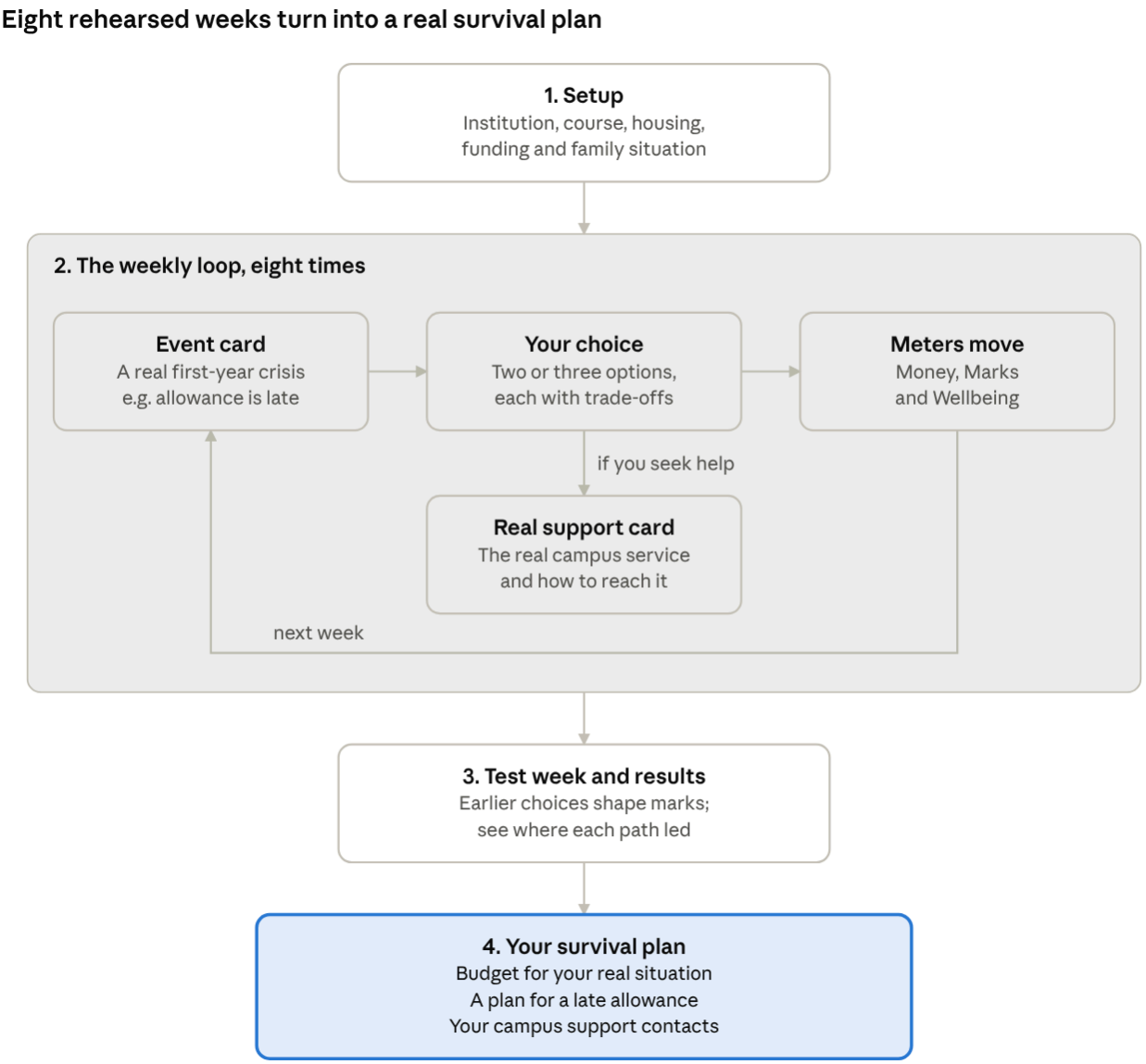
The main users are incoming first-years, played in the weeks between matric results and registration.

User	Who they are	What they get
Incoming first-years	Matriculants accepted to university or TVET, especially first-generation and NSFAS-funded students	A rehearsal of their real first semester and a personal survival plan
Parents and families	Families supporting a first-time student, often without university experience	A shared view of what the first year costs and demands
Student support offices	Student affairs, mentorship and first-year experience teams	A way to reach students before arrival, and insight into which crises worry them most

The pilot targets UP first-years at the Hatfield campus.

How it works

A player sets up their real situation, plays eight compressed weeks, and leaves with a survival plan, in about 15 minutes.



First Year Sim game flow · setup, 8-week loop, results, plan

Seeking help in any week reveals the real service at the player's campus, so students learn what exists before they need it.

Semester Forecast: how the app thinks

First Year Sim does not tell a fixed story. It reads the player's numbers and choices, spots patterns, and predicts where their real first semester is likely to go wrong.

Part	Input	What it predicts	Example
Money forecast	Allowance, rent, transport, family support	The week money is likely to run out, with and without a late allowance	"If your allowance is two weeks late, you'll run out of money in week 2."
Spending leaks	Friday nights, takeaways, clothes, subscriptions	How much small spends cost over the month	"Small spends cost you R900 this month. That's your missing rent money."
Behaviour pattern	The pattern of choices across weeks	Coping style, and the weeks it will hurt most	"You handled things alone 4 times out of 5. Your biggest risk is weeks 6 and 7."
Adaptive events	Live scores from all parts	Which event the player meets next	A player who joined nothing is more likely to meet the loneliness event

What the player sees at the end

> Highest risk: weeks 2 and 6. Main cause: money shortfall, then test-week stress. Your pattern: you avoid asking for help. Your plan: register for SNAPP and contact the SRC about NSFAS before week 2; book an Academic Success Coach before test week.

Today and after the pilot. In the hackathon build, the forecast combines a cash-flow calculation, a score for each choice across coping categories, and rules based on research into why students drop out. An optional AI model writes the plan in natural language when the phone is online, with a template fallback offline. In a pilot, anonymous, consented play data and first-term outcomes train a prediction model that becomes more accurate over time.

Example semester at the University of Pretoria

A BSc first-year from a village in Limpopo, on NSFAS, renting privately in Hatfield and without a laptop, plays eight compressed weeks. Each week brings one main event card with two or three choices. Some cards change depending on setup answers and earlier choices.

Week	Event	Choices	Real UP support revealed
1	First week: new to Pretoria and unsure which modules to take	Ask a senior; figure it out alone; book an advisor	Faculty Student Advisors and STARS first-year mentorship
2	Societies Day: a sports team, a gym deal with friends, or nothing	Join a team; sign the gym contract; skip it	Sports clubs and societies; how monthly debit orders add up
3	Allowance hasn't arrived and food is running out	Borrow from a friend; skip meals; get help	SNAPP food parcels, vouchers and meal allowances; SRC help with NSFAS issues
4	Family asks for R400	Send it; send part; explain you can't	Budget planning for family obligations
5	Chemistry test and maths assignment in the same week; harder without a laptop	Cram alone; drop one; get help	Academic Success Coaches; computer labs and any device support
6	Lonely, can't sleep, thinking of going home (more likely if the player joined nothing)	Push through; call home; reach out	Free, confidential Student Counselling Unit; societies; STARS mentors
7	Test week	Results depend on earlier choices	—
8	Results and what they mean	Review outcomes	Why key first-year modules and NSFAS progression rules matter

Every Friday: a quick choice to go out or stay in, plus small temptations such as takeaways or new clothes. Going out helps friendships and wellbeing but costs money; the game never judges, it shows the trade-off.

Any week: a surprise expense, such as a stolen phone or a required lab coat, tests whether the player kept a buffer. Players renting privately may also meet an accommodation scam: "cheap room near campus, pay the deposit today".

UP programme names come from UP's Youth Month overview, UP Student Affairs and a 2026 first-year account. UP's Director of Student Affairs has noted that failing subjects like maths and chemistry in first year can block progression to second year. Device

support and sports club details still need confirming with UP. Amounts in the game are example figures, clearly labelled.

Key features

The game is short, personal, and ends in something the student keeps.

- **Personal setup:** institution, campus, course, housing (residence, private, or commuting), funding source, where the student is coming from (same city, another city, or a village or rural area), whether they have a laptop, and whether family may need support. The semester is built from these answers.
- **Three meters:** Money, Marks and Wellbeing. Choices trade one against another, and neglect in early weeks makes later weeks harder.
- **Event cards:** realistic South African first-year crises, including late NSFAS allowances, Societies Day pressure, family requests, workload clashes, loneliness and stress.
- **Friday choices and small spends:** quick weekly decisions about going out, takeaways and clothes, which the forecast tracks.
- **Surprise expenses:** a random cost, like a stolen phone, that tests whether the player kept a buffer.
- **Real support cards:** when a player seeks help, the game shows the actual service at their institution, what it offers and how to reach it.
- **Semester Forecast:** a cash-flow forecast, spending-leak tracking, a coping-style profile, and events that adapt to how the player behaves.
- **Survival plan:** a one-page plan to keep, built around the forecast, with a budget, an emergency buffer target, a "my allowance is late" playbook, key first-term dates, and support contacts.
- **Replay:** different choices lead to different outcomes, so students can test strategies.
- **Languages:** English plus at least one local language in the pilot, such as isiZulu or Sepedi.
- **Share with family:** a simple summary parents can read, so the household plans together.

What exists and how we differ

The idea of a university transition game exists overseas; a South African version built around NSFAS, family obligations and local realities does not, as far as we found.

Existing option	What it does	Where it stops
One Hour Degree, University of Kent (UK)	Narrative game simulating a Kent degree, with knowledge and wellbeing points; won a 2020 Guardian award	Built for UK students and one university; no South African funding or living realities
Lincoln Island Project (UK research)	Testing a video game to support transition, covering finances, support services and campus navigation	Research stage, UK context
Orientation games, e.g. ANU Campus Quest	Scavenger-hunt apps for the first days on campus	Cover finding your way, not surviving the semester
Budgeting simulations, e.g. Stock-Trak	Practice managing income, bills and surprise events	General money practice, not university life
RU GAME, Rhodes University	Orientation theatre show depicting NSFAS delays, residence problems and academic pressure	A stage show, not a tool students can play or keep

Our difference: South African crises (NSFAS delays, black tax, accommodation scams, commutes, data costs), personalised to the student's real institution and funding, played before arrival, in local languages, ending in a usable survival plan. We will name One Hour Degree in the pitch ourselves.

Design principles

Four rules guide the content and build.

Real, not scary. Events reflect what actually happens, but every crisis comes with a way through. The goal is confidence, not fear of university.

No judgement. Choices like sending money home have no single right answer. The game shows consequences and options, never lectures.

Built for the least connected. A web app opened from a WhatsApp link, no download, no login, light on data, and usable on cheap phones. The game and the forecast run offline after the first load; only the optional AI-written plan needs a connection.

Privacy by default. No account and no personal identifiers. Setup answers stay on the device, and only anonymous, aggregate choices are counted if a university partner wants insight.

How First Year Sim meets the judging criteria

In round 1, judges play the app on their own, so it must explain itself, feel finished and work on any phone.

Criterion	How we meet it
Product	A complete eight-week game with no unfinished screens, tested on low-cost Android phones
Creativity	Rehearsing the first semester as a game, with South African crises and a forecast that adapts to each player
Usability	Short intro screens, one decision per screen, large buttons and clear meters
Sustainability	Content files make new institutions cheap to add; funded by universities, NSFAS and bursary providers
Impact	Targets first-year dropout, one of the largest losses in South African higher education
Presentation	Opens with a real student story, then the numbers, then a live play-through
Intelligence	Semester Forecast: cash-flow forecast, coping-style profile and adaptive events
Traction	Testing with real students before the hackathon (below)

Traction plan before the hackathon

- [ ] 30 to 50 matriculants or first-years play an early version
- [ ] Record completions, and how many discovered a support service they didn't know about
- [ ] Collect short quotes from current first-years about their first semester
- [ ] Ask UP student affairs, the SRC or a first-year mentor for an email or letter of support

Risks and how we address them

The questions judges are most likely to ask are whether a game changes behaviour and whether it already exists.

Risk	What could go wrong	How we address it
"It's just a game"	Students play once and forget	The survival plan and support contacts carry into real life; a pilot measures first-term outcomes
"This exists already"	A judge knows One Hour Degree	We name it first and show the South African, personalisation and forecast differences
Judges play alone	Judges get lost without the team present	Intro screens, clear next steps, and no unfinished features
Overclaiming intelligence	Predictions presented as more than they are	Explain that today's forecast is research-based rules, trained on real data after the pilot
AI needs internet	The AI-written plan fails offline	Game and forecast run offline; AI only polishes the plan when online
Outdated information	Contacts or programme names change	Content stored as editable files; each institution's details confirmed with its student affairs office
Scaring students off	Students decide university is too hard	Every crisis has a way through; the tone builds confidence
Low engagement	Students skip a 15-minute game	Short weeks, shareable results, and distribution through schools, NSFAS channels and university acceptance messages
Wrong figures	Allowance amounts or NSFAS rules quoted incorrectly	Use labelled example amounts; confirm current NSFAS rules before launch
Sensitive topics	Mental health events upset a player	Gentle wording and immediate access to real support contacts in those events

Technical approach and MVP scope

The hackathon build is one institution (UP), eight weeks, and ten to twelve event cards plus weekly mini-choices, playable start to finish.

Components

- **Front end:** a mobile-first web app (progressive web app) that works offline once loaded, and can be wrapped as an Android app if needed.
- **Game engine:** a simple state machine tracking the three meters, the current week, and flags set by earlier choices.
- **Forecast engine:** calculates weekly cash flow from setup answers, scores each choice against coping categories, and selects the next event from those scores.
- **Content files:** events, choices, outcomes and support cards stored as JSON per institution, so adding a university means adding content, not code.
- **Personalisation rules:** setup answers select which events appear and adjust costs, such as commute events only for commuters.
- **Survival plan generator:** builds the end-of-game plan from the forecast, downloadable as an image to save or share on WhatsApp. An optional AI model writes it in natural language when online.
- **Translations:** text stored per language, with English and one local language in the MVP.

Screens: intro (two or three screens), setup, weekly dashboard, event card, outcome, support card, semester forecast, survival plan.

In the MVP

- [ ] Intro screens that explain the app without the team present
- [ ] Setup flow with UP options, including hometown and laptop questions
- [ ] Eight-week game loop with three meters
- [ ] Ten to twelve UP event cards with branching and conditional outcomes
- [ ] Friday and small-spend mini-choices
- [ ] Random surprise expense
- [ ] Support cards for UP services
- [ ] Cash-flow and spending-leak forecast
- [ ] Coping-style scoring and adaptive event selection
- [ ] Semester forecast and survival plan screens, shareable as an image
- [ ] One local-language version

Left for later: more institutions including TVET colleges, parent mode, university dashboards, and real-semester check-ins.

Demo flow

There are two judging rounds, and each needs its own experience.

Round 1: judges play alone. Judges open the app cold. Short intro screens state the problem in one line, explain the game, and start the semester. The full eight weeks take about 15 minutes, and every path ends at the forecast and survival plan, so judges see the intelligence without being told about it.

Round 2: live pitch (about three minutes)

1. **The hook (20 s).** "About 620,000 NSFAS-funded first years in 2023. Only about 130,000 were still funded a year later."
2. **Setup (20 s).** The judge picks UP, BSc, private room in Hatfield, NSFAS. The money forecast appears straight away.
3. **Week 2 (40 s).** Their allowance is late. The judge chooses; the meters move; the SNAPP support card appears.
4. **Week 4 (30 s).** Family asks for R400. There is no single right answer, only trade-offs.
5. **Week 6 (30 s).** Wellbeing drops, and the free Student Counselling Unit card appears.
6. **The forecast (40 s).** The Semester Forecast names the judge's riskiest weeks and coping style, then the survival plan is shared to WhatsApp.

Close on the line: "Every first-year crisis is predictable. First Year Sim lets students rehearse it, and predicts which ones will hit them hardest."

Impact, sustainability and roadmap

Success means more first-years using support early and finishing their first term in good standing.

How we measure it

- Number of incoming first-years who complete the game and save their survival plan
- Share who say they know where to get help with food, money and mental health
- In a pilot: use of support services in the first term by players compared with non-players
- In a pilot: first-term pass rates and continuation of funding for players compared with non-players

Who pays

Students never pay. Likely funders are universities' student affairs and first-year experience budgets, NSFAS or the Department of Higher Education for national rollout, and bursary providers and corporate social investment programmes that already fund student success.

Roadmap

1. **Hackathon:** UP version, eight weeks, Semester Forecast, one local language.
2. **Pilot:** partner with UP student affairs; send the game with acceptance messages; measure first-term outcomes; use anonymous, consented play data to train the forecast model.
3. **Expand:** add more universities and TVET colleges through content files; add parent mode; add an international student path covering study permits and funding without NSFAS.
4. **Real-semester companion:** optional weekly check-ins during the actual first term, where the forecast flags trouble early and points students to support.

To confirm before pitching

- [ ] Whether the organisers need an installable Android app or accept a web link
- [ ] Current UP support programme names and contact details with the Department of Student Affairs
- [ ] Current NSFAS allowance timing and academic progression rules
- [ ] Realistic example costs for Hatfield rent and transport